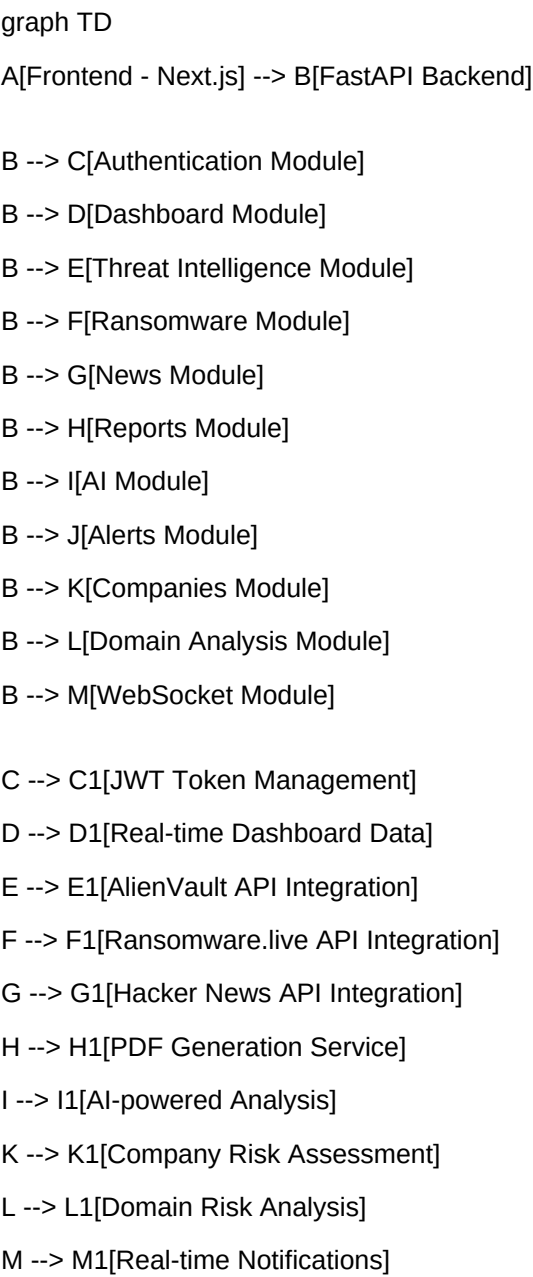


# VAJRA Security Platform - API Flow Diagram

## Overview

This document provides a comprehensive flow diagram of all API endpoints in the VAJRA Security Platform, showing their purposes and data flow.

## API Architecture Flow



# Detailed API Endpoints

## 1. Authentication Module (`/api/auth`)

**\*\*Purpose\*\*:** User authentication and authorization

POST /api/auth/register

■■■■ Purpose: Register new user

■■■■ Input: UserCreate (email, name, password)

■■■■ Output: UserResponse (id, email, name, role, is\_active)

■■■■ Database: Creates user in PostgreSQL

POST /api/auth/login

■■■■ Purpose: User login and token generation

■■■■ Input: UserLogin (email, password)

■■■■ Output: Token (access\_token, refresh\_token, user)

■■■■ Security: JWT tokens for authentication

POST /api/auth/refresh

■■■■ Purpose: Refresh access token

■■■■ Input: TokenRefresh (refresh\_token)

■■■■ Output: Token (new access\_token, refresh\_token, user)

■■■■ Security: Token validation and renewal

## 2. Dashboard Module (`/api/dashboard`)

**\*\*Purpose\*\*:** Real-time dashboard metrics and visualization data

GET /api/dashboard/summary

■■■■ Purpose: Get overall dashboard summary

■■■■ Output: DashboardSummary (total\_attacks, active\_threat\_actors, critical\_attacks, last\_updated)

■■■■ Data Source: AlienVault API

■■■■ Usage: Main dashboard cards

GET /api/dashboard/alerts

■■■■ Purpose: Get recent critical alerts

■■■■ Output: List[Alert] (id, title, severity, description, time, source)

■■■ Data Source: Mock data + AlienVault alerts

■■■ Usage: Critical alerts component

GET /api/dashboard/attack-map

■■■ Purpose: Get attack map visualization data

■■■ Output: List[AttackMapData] (source, target, coordinates, count)

■■■ Data Source: Mock geographic attack data

■■■ Usage: Attack map visualization

### 3. Threat Intelligence Module (`/api/threat-intelligence`)

**\*\*Purpose:** Comprehensive threat intelligence data and analysis

GET /api/threat-intelligence

■■■ Purpose: Get overall threat intelligence metrics

■■■ Output: ThreatIntelligence (score, threatActors, malwareFamilies, iocCount)

■■■ Data Source: AlienVault API

■■■ Usage: Threat intelligence dashboard

GET /api/threat-intelligence/trend

■■■ Purpose: Get threat trend data over time

■■■ Output: List[ThreatTrend] (date, score)

■■■ Data Source: Mock historical data

■■■ Usage: Trend charts and graphs

GET /api/threat-intelligence/actors

■■■ Purpose: Get threat actor information

■■■ Output: List of threat actors with details

■■■ Usage: Threat actor tracking

GET /api/threat-intelligence/industries

■■■ Purpose: Get industry-specific threat data

■■■ Output: Industry threat statistics

■■■ Usage: Industry risk assessment

### 4. Ransomware Module (`/api/ransomware`)

**\*\*Purpose:** Ransomware attack tracking and analysis

GET /api/ransomware

■■■ Purpose: Get recent ransomware incidents

■■■ Output: List[RansomwareIncident] (id, group, target, country, published, impact, status, description)

■■■ Data Source: Ransomware.live API

■■■ Usage: Ransomware incident tracking

GET /api/ransomware/stats

■■■ Purpose: Get ransomware statistics

■■■ Output: RansomwareStats (groupsCount, overallVictims, victimsThisYear, victimsThisMonth, trends)

■■■ Data Source: Ransomware.live API

■■■ Usage: Ransomware statistics dashboard

GET /api/ransomware/group/{group\_name}

■■■ Purpose: Get incidents for specific ransomware group

■■■ Output: List[RansomwareIncident] for specific group

■■■ Data Source: Ransomware.live API

■■■ Usage: Group-specific analysis

## 5. News Module (`/api/news`)

**\*\*Purpose:** Live cyber threat news from Hacker News

GET /api/news

■■■ Purpose: Get cybersecurity-related news

■■■ Output: List of news stories (title, url, time\_ago, score)

■■■ Data Source: Hacker News API

■■■ Filtering: Cybersecurity keyword filtering

■■■ Usage: Live cyber threat news component

## 6. Reports Module (`/api/reports`)

**\*\*Purpose:** PDF report generation for various modules

GET /api/reports/threat-intelligence

■■■ Purpose: Generate threat intelligence PDF report

■■■ Output: PDF file (threat\_intelligence\_report.pdf)

■■■ Service: PDF generation with black-and-white styling

■■■ Usage: Download threat intelligence report

GET /api/reports/ransomware

■■■ Purpose: Generate ransomware PDF report

■■■ Output: PDF file (ransomware\_report.pdf)

■■■ Service: PDF generation with black-and-white styling

■■■ Usage: Download ransomware report

GET /api/reports/global-attacks

■■■ Purpose: Generate global attacks PDF report

■■■ Output: PDF file (global\_attacks\_report.pdf)

■■■ Service: PDF generation with black-and-white styling

■■■ Usage: Download global attacks report

GET /api/reports/company/{company\_id}

■■■ Purpose: Generate company-specific PDF report

■■■ Output: PDF file ({company\_name}\_report.pdf)

■■■ Database: Company, CompanyThreat, CompanyRiskAssessment tables

■■■ Usage: Download company risk report

GET /api/reports/executive

■■■ Purpose: Generate executive summary PDF report

■■■ Output: PDF file (executive\_summary\_report.pdf)

■■■ Service: PDF generation with black-and-white styling

■■■ Usage: Download executive summary

GET /api/reports/comprehensive

■■■ Purpose: Generate comprehensive PDF with all modules

■■■ Output: PDF file (vajra\_comprehensive\_report.pdf)

■■■ Service: Combined PDF generation

■■■ Usage: Download comprehensive report

GET /api/reports/project-overview

■■■ Purpose: Generate project overview PDF with ransomware focus

■■■ Output: PDF file (vajra\_project\_overview.pdf)

■■■ Service: Project overview PDF generation

■■■ Usage: Download project overview report

## 7. AI Module (`/api/ai`)

**\*\*Purpose:** AI-powered threat analysis and predictions

POST /api/ai/analyze

■■■■ Purpose: AI-powered threat analysis

■■■■ Input: Threat data for analysis

■■■■ Output: AI analysis results

■■■■ Usage: Advanced threat analysis

GET /api/ai/predictions

■■■■ Purpose: Get AI threat predictions

■■■■ Output: Predictive threat data

■■■■ Usage: Threat forecasting

## 8. Alerts Module (`/api/alerts`)

**\*\*Purpose:** Alert management and notifications

GET /api/alerts

■■■■ Purpose: Get all alerts

■■■■ Output: List of alerts

■■■■ Database: Alert table

■■■■ Usage: Alert management

POST /api/alerts

■■■■ Purpose: Create new alert

■■■■ Input: Alert data

■■■■ Database: Creates alert in PostgreSQL

■■■■ Usage: Alert creation

## 9. Companies Module (`/api/companies`)

**\*\*Purpose:** Company management and risk assessment

GET /api/companies

■■■■ Purpose: Get all companies

■■■■ Output: List[Company]

■■■■ Database: Company table

■■■ Usage: Company list

GET /api/companies/{company\_id}

■■■ Purpose: Get specific company details

■■■ Output: Company with risk assessment

■■■ Database: Company, CompanyRiskAssessment tables

■■■ Usage: Company detail view

POST /api/companies

■■■ Purpose: Add new company for monitoring

■■■ Input: Company data

■■■ Database: Creates company in PostgreSQL

■■■ Usage: Company onboarding

## 10. Domain Analysis Module (`/api/domain-risk`, `/api/domain-analysis`)

**\*\*Purpose:** Domain risk assessment and analysis

GET /api/domain-risk/{domain}

■■■ Purpose: Get domain risk assessment

■■■ Output: Domain risk data

■■■ Data Source: Domain analysis services

■■■ Usage: Domain risk evaluation

GET /api/domain-analysis/{domain}

■■■ Purpose: Detailed domain analysis

■■■ Output: Comprehensive domain analysis

■■■ Usage: Deep domain inspection

## 11. Notifications Module (`/api`)

**\*\*Purpose:** User notifications and alerts

GET /api/notifications

■■■ Purpose: Get user notifications

■■■ Output: List of notifications

■■■ Usage: Notification center

PUT /api/notifications/{notification\_id}

■■■ Purpose: Mark notification as read

■■■ Usage: Notification management

## 12. Admin Module (`/admin`)

**\*\*Purpose:** Administrative functions

## GET /admin/users

■■■ Purpose: Get all users (admin only)

■■■ Usage: User management

## GET /admin/stats

■■■ Purpose: Get platform statistics

■■■ Usage: Platform monitoring

### 13. WebSocket Module (`/ws`)

**\*\*Purpose:** Real-time data streaming

WebSocket /ws

■■■ Purpose: Real-time threat data streaming

■■■ Events: threat\_updates, alert\_notifications, dashboard\_refresh

■■■ Usage: Live data updates without polling

## Data Flow Architecture

[illegible]

## ■ Frontend (Next.js) ■

**[REDACTED]**

■ ■ Dashboard ■ ■ Threat ■ ■ Ransomware ■ ■ Reports ■ ■

[illegible]

**[REDACTED]**

4444

[illegible]

■

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## AlienVault OTX API

- **Purpose**: Threat intelligence data
- **Endpoints Used**:
  - Threat intelligence metrics
  - IOC data
  - Alert data
- **Service**: `services/alienvault\_service.py`

## Ransomware.live API

- **Purpose**: Ransomware attack tracking
- **Endpoints Used**:
  - Recent ransomware attacks
  - Ransomware group statistics
  - Group-specific incidents
- **Service**: `services/ransomware\_service.py`

## Hacker News API

- **Purpose**: Cyber threat news
- **Endpoints Used**:
  - Top stories
  - New stories
  - Story details
- **Service**: `services/hackernews\_service.py`

## Database Schema

### Tables

- **users**: User authentication and profile data
- **companies**: Company information for monitoring
- **company\_threats**: Threat data associated with companies
- **company\_risk\_assessments**: Risk assessment results
- **alerts**: Security alerts and notifications
- **threat\_actors**: Threat actor information

- **industries**: Industry-specific threat data

## Security Features

- **JWT Authentication**: Token-based authentication
- **CORS**: Cross-origin resource sharing enabled
- **Rate Limiting**: API rate limiting middleware
- **Security Headers**: HTTP security headers
- **Password Hashing**: Secure password storage

## WebSocket Events

- **threat\_updates**: Real-time threat intelligence updates
- **alert\_notifications**: New alert notifications
- **dashboard\_refresh**: Dashboard data refresh triggers
- **ransomware\_updates**: New ransomware incident alerts

## File Downloads

Static file downloads available:

- PowerPoint presentations (.pptx)
- Pre-generated PDF reports
- Project documentation

## Summary

The VAJRA Security Platform API consists of 13 main modules with over 30 endpoints, providing comprehensive threat intelligence, ransomware tracking, company risk assessment, and reporting capabilities. The architecture follows a clean separation of concerns with dedicated service layers for external API integrations and database operations.